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Out-of-distribution detection-assisted trustworthy machinery fault diagnosis approach with uncertainty-aware deep ensembles

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ABSTRACT

Recent intelligent fault diagnosis technologies can effectively identify the machinery health condition, while they are learnt based on a closed-world assumption, i.e., the training and testing data follow independently identically distribution (IID). However, in real-world diagnosis, the monitored samples are often from unknown distributions, such as unseen machine faults, leading to an out-of-distribution (OOD) problem. This is a challenging issue that may induce the model to produce unreliable and unsafe decision for unforeseen machine data. To tackle this problem, a novel OOD detection-assisted trustworthy machinery fault diagnosis approach is developed to enhance the reliability and safety of intelligent models. First, multiple deep neural networks are integrated to establish an ensemble diagnosis system, called deep ensembles. Then, the trustworthy analysis with uncertainty-aware deep ensembles is conducted to detect the OOD samples and issue the warnings for the potential untrustworthy diagnosis. A selection criterion of uncertainty threshold is given. Finally, the trustworthy decisions are achieved by comprehensively considering the deep ensembles' prediction and uncertainty. The proposed trustworthy fault diagnosis approach is validated in two case studies, exhibiting significant advantages for diagnosing OOD samples.

1. Introduction

Rotating machinery is widely employed in different critical engineering fields, such as manufacturing system, rail transportation, power generation, etc, and plays a pivotal role in economic production. The surveillance and diagnosis of possible failures in industrial machinery equipment have important practical needs to ensure its reliable and continuous operation, reduce the operating costs as well as avoiding major accidents [1,2]. The development of artificial intelligence has provided new ideas and opportunities for machinery fault diagnosis [3]. Especially, the deep learning-based fault diagnosis in a data-driven framework can largely reduce the reliance on experts knowledge and engineering experience, forming end-to-end diagnosis system [4–6].

Existing deep learning methods necessitate various types of data (including healthy and faulty data) to train the high-accuracy diagnosis model. However, the samples and corresponding labels available for model learning are limited in practical industrial scenarios [7,8]. It is difficult to obtain sufficient training data with different faults, as it takes a period of time from healthy condition to early machine failure. The diagnostic models are generally trained on the limited data, and thus are only applicable to the known distribution inputs, also referred to in-distribution (ID) inputs [9]. Due to the complex fault mechanism and varying application environments, the real monitored data are likely drawn from the different distributions with respect to training data, lead to an out-of-distribution (OOD) problem [10,11]. For instance, there are a number of possible faults in a mechanical system, and the training data can only cover finite types. In this case, the trained models cannot intelligently identify the unseen faults, and will make the false and overconfident diagnosis decision silently [12,13]. This is dangerous in safety-critical applications. It is significant to consider the trustworthiness of the outputs of intelligent diagnosis model.

Recently, the domain shift is another often discussed topic in machinery fault diagnosis, which also pays attention to the distribution discrepancy problems between source and target domains. The corresponding transfer fault diagnosis framework, mainly including parameter transfer and domain adaptation, is proposed to address the domain shift, such that the transferred model can generalize to the target distributions, which are different from the ones of training data [14–18]. Despite its success, the transfer fault diagnosis needs certain prior knowledge on the target data to induce the domain shift minimization between source and target domains. And thus, the transferred model

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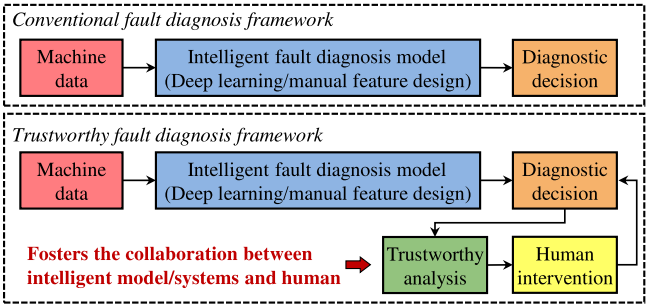

Fig. 1. The illustration of trustworthy fault diagnosis framework.

is still limited to a specific target data distribution, showing its shortcomings in the practical applications. The unreliability problem of the models' outputs in an OOD scenario have not been completely solved in the transfer diagnosis approaches.

To form a trustable fault diagnosis approach, the intelligent models should not only focus on the diagnostic decisions and accuracy, but also evaluate their trustworthiness, which are only neglected in the black-box models. An additional stage, namely the trustworthy analysis, should be considered to detect the occurrence of previously unknown data, so as to issue the warning for the potential untrustworthy diagnosis and turn to human evaluation [19]. The trustworthy analysis is conducive to foster the collaboration between intelligent models and human, and thus improves the models' reliability and safety. Fig. 1 illustrates the difference between conventional and trustworthy fault diagnosis frameworks. OOD detection holds the potential to implement the trustworthy analysis, which aims to assess whether the monitored data is drawn from the same distribution as the training data, i.e., ID data, or comes from a different domain, i.e., OOD data. Identifying and addressing these OOD data is crucial, as the fault diagnosis model should be substantially uncertain about data outside of the known/training distribution, and will give untrustworthy diagnostic decisions. OOD detection has attracted more and more attention in several safety-critical fields, such as autonomous driving [20] and healthcare [21–23], but is relatively unexplored in machinery fault diagnosis. The source of OOD samples in machinery fault diagnosis is broad, mainly including the previously unseen classes (namely unseen machine health conditions/faults), the distribution shift caused by the unseen working conditions, as well as the unseen noise environments. This study will focus on the most challenging unseen fault detection and diagnosis to overcome this restriction of closed-set assumption in the data-driven methods, especially the deep learning methods.

The state-of-the-art OOD detection methods in deep learning mainly include softmax output-based methods, reconstruction-based methods, uncertainty-based methods, etc [24,25]. The presented method in this study belongs to the category of uncertainty-based methods. Uncertainty-based OOD detection concentrates on developing highly accurate approaches to quantify the predictive uncertainty of OOD samples [26]. There is always a predictive uncertainty along with the decision made by fault diagnosis model. During decision-making, a high confidence level will be given for the known ID data, whereas a low confidence level will be assigned for OOD data, because the diagnostic information is unknown to the model. The uncertainty is effective to reflect the confidence level, that can be exploited to detect ID or OOD samples. Unfortunately, deep learning is generally poor at manifesting the predictive uncertainty in diagnostic results. The commonly used deep learning models are inherently deterministic, and often make overconfident decisions [27,28]. This phenomenon makes it difficult to separate the OOD data from the ID data. To overcome this limitation, this study investigates the OOD detection with the ensembles of base deep neural networks. It has long been seen that ensemble learning outperforms the single model in terms of diagnostic performance and generalization ability. Recent study further reveals that ensemble of base learners will possess stronger capacity to quantify the uncertainty of diagnosis results [29,30].

In this study, a novel OOD detection-assisted trustworthy machinery fault diagnosis approach is proposed to flag and reject the OOD samples for the trained model, so as to avoid the untrustworthy diagnosis decisions. Specifically, we present a uncertainty-aware diagnosis model with the ensemble of multiple deep base learners (DBLs) to detect the unknowns during decision-making. With the proposed approach, we aim to explore the significant but less reported trustworthy fault diagnosis, establish an automatic detection procedure of unseen faults, as well as providing guidance for potential expert intervention in safety-critical applications. The main contributions are twofold:

(1) A potentially significant and new area, i.e., trustworthy fault diagnosis, is investigated. The OOD detection is implemented to guarantee the trustworthy diagnosis of ID data and tackle the challenges of unseen faults diagnosis in practical industry applications, which relaxes the closed-set assumption in the most of intelligent fault diagnosis methods.

(2) A simple and efficient ensemble of DBLs is described, that is capable of not only accurately diagnosing the health conditions of ID samples, but also estimating the predictive uncertainty during diagnosis. A proper criterion is given to adaptively determine the uncertainty threshold, such that the OOD samples with unseen faults can be automatically identified.

The rest of the paper is laid out as follows. The basic models and proposed methodology are summarized in Section 2. Then, Section 3 introduces two case studies to demonstrate the proposed approach. Finally, Section 4 concludes the results and future work.

2. Methodology

2.1. Problem formulation

The trustworthiness in intelligent fault diagnosis is investigated in this study. Firstly, the ID training data with a specific distribution can be obtained for intelligent model establishment. Let $\mathcal{X}$ and $\mathcal{Y}$ denote the ID input and output label space. Let $\mathcal{P}$ be the distribution over $\mathcal{X} \times \mathcal{Y}$. $f : \mathcal{X} \rightarrow \mathbb{R}^{|\mathcal{Y}|}$ represents the trained diagnosis model with ID data. Deep learning models have proved to be effective to achieve highly accurate diagnosis results for ID data, whose distribution is denoted as $\mathcal{D}_{in}$. Consequently, the diagnosis results given by these deep models are assumed to be trustable.

In the real deployment scenarios, the trained model f inevitably encounters the OOD data, whose distribution $\mathcal{D}_{out}$ is different from the $\mathcal{D}_{in}$. This difference is reflected both in semantic shift and covariate shift. Detecting these OOD inputs is paramount in forming a trustworthy fault diagnosis framework, as well as enhancing the reliability and safety of intelligent models. Given a testing sample x, this study aims to establish a function F:

$$F(x; f) = \begin{cases} trustworthy & if \quad x \sim \mathcal{D}_{in} \\ untrustworthy & if \quad x \sim \mathcal{D}_{out} \end{cases} . \tag{1}$$

2.2. Deep base learners

Different kinds of deep neural networks generally have different architectures and learning mechanisms to capture the feature representation of machine health conditions. In this study, five neural networks in the family of convolutional neural network (CNN) are designed as the DBLs of deep ensemble, which aims to induce diversity for achieving more accurate uncertainty estimates. This section gives the brief descriptions of the designed DBLs.

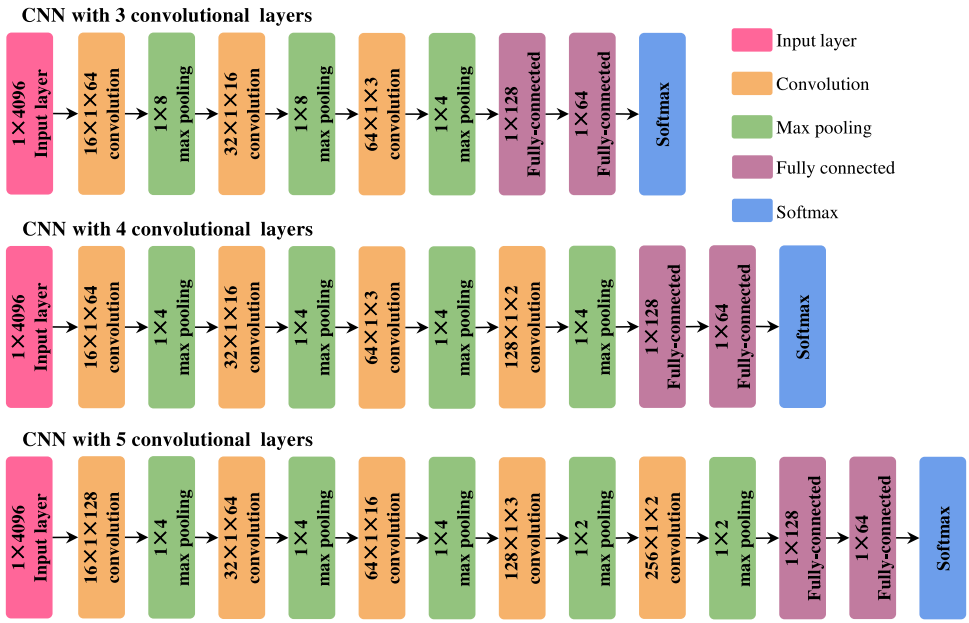

Fig. 2. The architecture of three standard CNNs.

2.2.1. CNN

As one of the most popular deep neural networks, CNNs are selected as the standard DBLs in this study. The convolutional layer firstly convolves the raw inputs using numerous learnable kernels. Then, the outputs are processed by the nonlinear activation function. Generally, each convolution operation is followed by pooling layer, which aims to reduce the dimension of features and increase the receptive field. With the stacked convolution and pooling operations, a deep feature extractor is constructed. The fully-connected layers further map the extracted features into the classes of machine health conditions. The softamx layer is appended at the end of network. Supposing that C known machine health conditions need to be diagnosed, the final predicted probability in softmax layer can be formulated as:

$$\begin{bmatrix} p(\hat{y}=1|x,\theta) \\ p(\hat{y}=2|x,\theta) \\ \cdots \\ p(\hat{y}=C|x,\theta) \end{bmatrix} = \frac{1}{\sum_{i=1}^{C}\exp(o_i)} \begin{bmatrix} \exp(o_1) \\ \exp(o_2) \\ \cdots \\ \exp(o_C) \end{bmatrix}, \tag{2}$$

where x is the input sample, $\hat{y}$ is the predicted probability, θ is the network parameters, o_i is the output of the ith neuron in softamx layer.

In machine fault diagnosis, the commonly used architectures generally include three, four or five convolutional layers [31–34]. Accordingly, the details of three designed CNNs are shown in Fig. 2. All the models adopt the one-dimensional convolutional kernels to form an end-to-end learning manner.

2.2.2. ResNet

The proposal of ResNet is a milestone event in the history of CNN [35]. Since AlexNet, the advanced CNN architectures have been evolving, becoming deeper and deeper. The AlexNet [36], VGG network [37] and GoogleNet [38] have 5, 19 and 22 layers, respectively. However, with the increase of network depth, the models are difficult to train due to the vanishing gradient problem. To handle this problem, a novel identity shortcut connection is used in ResNet. The identity shortcut connection can permit the gradient to flow through a alternative way, and sikp the layers that negatively affect performance. Fig. 3 presents the architecture of designed ResNet in this study. It is a variant of the ResNet18 architecture [35]. The one-dimensional convolutional kernels are used to adapt to the raw machine data.

2.2.3. Inception network

Inception block is proposed in GoogleNet [38]. Generally, choosing the proper kernel size for the convolution process becomes difficult due to the complex components in the machine fault signal. For the globally distributed components with low frequency, a larger kernel is desired, while a smaller kernel is preferred for the high frequency components that are locally dispersed. Now in inception block, multiple kernels with different sizes are integrated to perform convolutional operations on the signals and achieve multi-scale feature extraction. The outputs of different convolution kernels are finally cascaded and transferred to the next layer. By using such inception block, the fault features with different scales can be simultaneously extracted by the kernels with different sizes. Besides, the difficulties of the model training with deeper architecture can be solved by reducing the gradient decline in backpropagation through parallel construction.

Fig. 4 shows the architecture of designed inception network in this work. First, a wide convolutional block with large kernels (1×128) is used. Then, two inception blocks are stacked. The kernels with 1×31 and 1×17 are used in the first inception block, and the kernels with 1×5 and 1×3 are used in the second inception block. Finally, the softmax layer gives the predicted probability of machine health conditions.

2.2.4. Configurations of hyper-parameters

Overall, the network architectures follow the following principles. The first convolutional layer uses the large kernels to capture low-frequency fault features and suppress high-frequency noise. The subsequent convolutional layers use smaller kernels, while the number of channels increase to learn abstract feature representation. The other hyper-parameters during model training are listed as follows. The optimization method is stochastic gradient descent with the learning rate of 0.001. The batch size is 32. The number of training epochs is determined by the early stopping criterion (patience 10). All the DBLs are trained in the PyTorch framework. The other parameters in network architectures and model training are all set to default values in PyTorch framework. For instance, the stride is default 1 in the absence of notes, and the padding is default 0 for these DBLs.

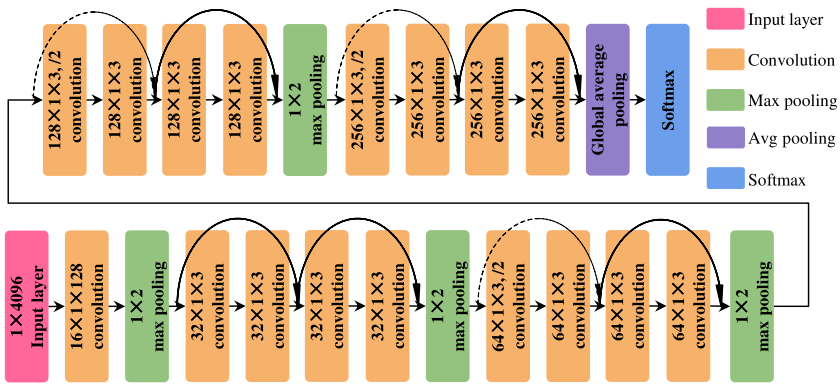

Fig. 3. The architecture of designed ResNet.

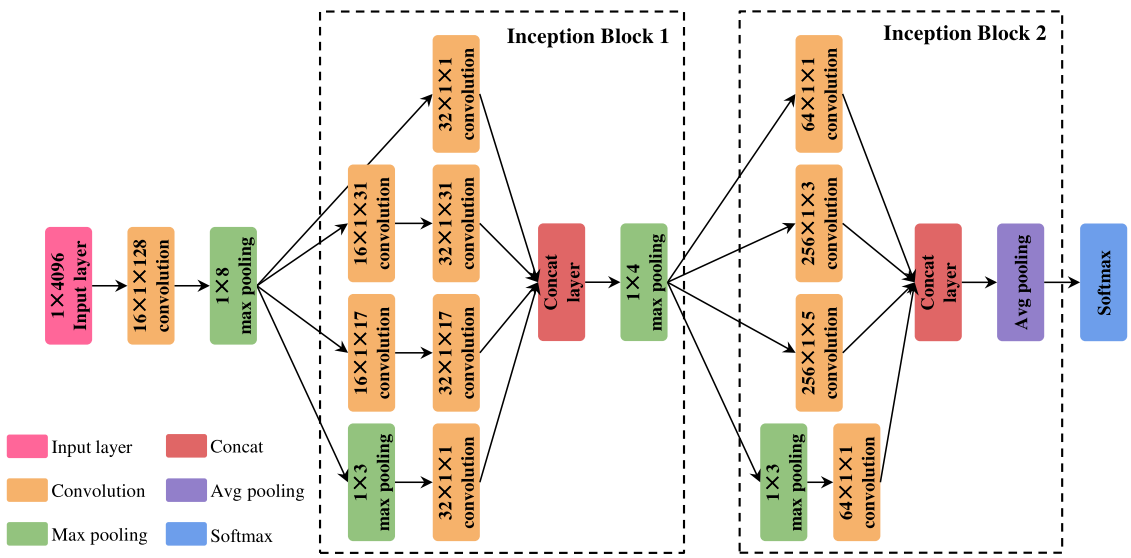

Fig. 4. The architecture of designed inception network.

2.3. Out-of-distribution detection with deep ensembles

2.3.1. Deep ensembles

On the basis of the described DBLs, the next step is to construct an ensemble diagnosis system, called deep ensembles. Although the individual DBLs have already been proved to be effective in terms of adaptive feature learning and diagnosis, the deep ensembles aims to handle the accurate diagnosis and its trustworthy analysis with uncertainty estimation at the same time. The presented deep ensembles is a kind of simple and efficient ensemble. It independently trains each DBL in parallel. This kind of ensemble is similar to the random forest approach. The output of diagnosis result is the averaging of all the individual DBLs. In essence, the deep ensembles can be regarded as a uniformly-weighted mixture diagnosis system. The predicted probability of machine health conditions is denoted as Eq. (3):

$$p(\hat{y}|x) = \frac{1}{M}\sum_{m=1}^{M} p(\hat{y}|x,\theta_m), \tag{3}$$

where $p(y|x,\theta)$ is the predicted probability distribution over machine health conditions by each DBL, θ_m is the network parameters of mth DBL, and the M is 5 in this study.

Note that the individual DBLs are all trained with the entire training data, as the deep learning models typically requires more data to get superior performance. This is different to the bagging strategy in traditional ensemble learning, which is used to generate diversity for the ensemble members [39]. In the presented deep ensembles, the different initialization of the model architectures and learning mechanisms can induce the sufficient diversity to the DBLs. As a consequence, the ensemble diagnosis system is capable of reflecting the uncertainty of diagnostic results.

2.3.2. Uncertainty estimation

Most of reported studies in machine fault diagnosis only focus on the accuracy, while ignore the trustworthiness of diagnostic decision. In safety-critical applications, it is significant to conduct the additional trustworthy analysis about the outputs given by these intelligent black-box models. Once the intelligent models handle the OOD samples, especially unseen fault samples, it is dangerous to adopt the untrustworthy results. Uncertainty is an efficient measure to tell human experts when the models' decisions can be trusted and how certain are the models. Therefore, this study considers the trustworthy machine fault diagnosis with uncertainty estimation, so that automatically detecting the OOD inputs and emitting the warnings. The overall framework is shown in Fig. 5.

In deep learning, the predicted softmax probability is one way to reflect the uncertainty of diagnostic results. The softmax probability that assigns all the probability mass to a specific category indicates the model is confident about its output decision. On the contrary, the softmax probability that distributes the probability mass evenly across all machine health categories suggests that the output decision is unconfident. For the samples from known machine health conditions,

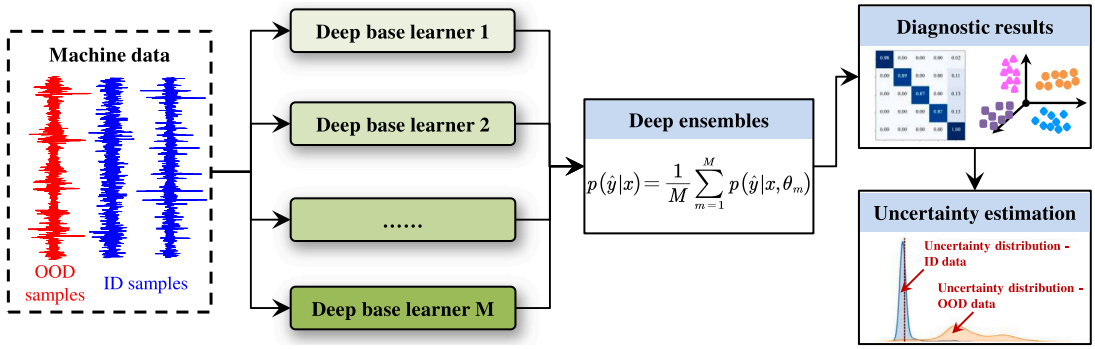

Fig. 5. The trustworthy machine fault diagnosis framework with uncertainty estimation.

the diagnostic model is expected to be confident and place the majority of probability mass to the true category. For the unseen fault samples, the fault features and diagnostic knowledge is unknown to the trained model, and thus the output of softmax probability should be unconfident. The entropy of softmax probability distribution can describe this confidence and be calculated as the quantitative uncertainty estimates. The entropy of softmax probability distribution can be calculated as:

$$\begin{aligned}\mathbb{H}[\hat{y}|x] &= -\sum_{c=1}^{C} p(\hat{y}=c|x)\times\log p(\hat{y}=c|x)\\ &= -\sum_{c=1}^{C}\left(\frac{1}{M}\sum_{m=1}^{M} p(\hat{y}=c|x,\theta_m)\right)\log\left(\frac{1}{M}\sum_{m=1}^{M} p(\hat{y}=c|x,\theta_m)\right),\end{aligned} \tag{4}$$

Note that the individual DBLs can also express its uncertainty through the entropy of softmax probability distribution. However, the individual DBLs tend to make overconfident diagnosis on unseen fault samples, which cannot be used for uncertainty estimates [27]. Essentially, the uncertainty sources can be categorized into two aspects: aleatoric uncertainty and epistemic uncertainty [26,40,41]. The aleatoric uncertainty is induced by the inherent randomness in the data. The epistemic uncertainty is referred to model's uncertainty caused by insufficient knowledge. In the individual DBLs, the aleatoric uncertainty can be learnt directly, whereas the epistemic uncertainty is much more challenging to estimate, due to the deterministic parameters and architectures. To answer when the inputs are unknown for the model, the ensemble diagnosis system containing the members with different initialization is necessary. Different members are likely to make different decisions for the unseen input from $\mathcal{D}_{out}$, and the epistemic uncertainty can be approximated in this manner.

Here, a hypothetical instance with two categories classification is presented. Given the ID input x_{in} from the $\mathcal{D}_{in}$, all the five individual DBLs and deep ensembles can accurately diagnose it, and the softmax output of the six models can be obtained:

$$\begin{aligned}p(\hat{y}|x_{in},\theta_1) &= [0.0, 1.0]\\ p(\hat{y}|x_{in},\theta_2) &= [0.0, 1.0]\\ p(\hat{y}|x_{in},\theta_3) &= [0.0, 1.0]\\ p(\hat{y}|x_{in},\theta_4) &= [0.0, 1.0]\\ p(\hat{y}|x_{in},\theta_5) &= [0.0, 1.0]\\ p(\hat{y}|x_{in}) &= [0.0, 1.0]\end{aligned} \tag{5}$$

According to Eq. (4), the uncertainties of all the six models are zero, indicating the trustworthy diagnosis results.

When handling the OOD input x_{out} from $\mathcal{D}_{out}$, such as the unseen fault sample, the five individual DBLs are confused and cannot make accurate diagnosis. Even if the individual DBL yields overconfident diagnosis, the disagreement among them can also express the epistemic uncertainty. For instance, we can obtain:

$$\begin{aligned}p(\hat{y}|x_{out},\theta_1) &= [1.0, 0.0]\\ p(\hat{y}|x_{out},\theta_2) &= [0.0, 1.0]\\ p(\hat{y}|x_{out},\theta_3) &= [1.0, 0.0]\\ p(\hat{y}|x_{out},\theta_4) &= [0.0, 1.0]\\ p(\hat{y}|x_{out},\theta_5) &= [1.0, 0.0]\\ p(\hat{y}|x_{out}) &= [3/5, 2/5]\end{aligned} \tag{6}$$

where the individual DBLs' uncertainties are zero, whereas the one of deep ensemble is a large value, namely $-3/5\log(3/5) - 2/5\log(2/5)$. Intuitively, the effectiveness of deep ensembles in terms of uncertainty estimation is clearly illustrated.

2.3.3. Selection of uncertainty threshold

Uncertainty can tell human experts whether the diagnostic decisions are trustworthy or untrustworthy. The former generally yields lower uncertainty with high confidence on the decisions, while the later has a higher uncertainty, reflecting the model's unknowns during diagnosis. To automatically judge the trustworthiness of diagnostic decisions, a proper uncertainty threshold need to be determined to detect the OOD samples. A desired uncertainty threshold should be larger than most of ones of ID samples, but lower than the ones of OOD samples. As the OOD samples are unforeseen during model training, the uncertainty threshold can only be determined within the validation set of ID samples. In this study, the inter-quartile range (IQR) proximity rule is used to find the upper bound of uncertainty values in ID samples, which is set to the threshold. The uncertainty threshold is calculated as follows:

$$\begin{aligned}IQR &= Q3 - Q1\\ \text{Upper bound} &= Q3 + 1.5IQR,\end{aligned} \tag{7}$$

where the $Q1$ and $Q3$ are the first quartile and the third quartile of the uncertainty values in ID validation set, respectively.

The IQR proximity rule is helpful to find the patterns and outliers of uncertainty values in ID validation set. This uncertainty threshold can ensure most of the ID samples are considered trustworthy, while detect the outliers of ID samples as well as the OOD samples with higher uncertainty.

2.4. Proposed trustworthy fault diagnosis approach

To detect the OOD samples and enhance the reliability and safety of intelligent diagnosis decisions, a trustworthy fault diagnosis approach with uncertainty-aware deep ensemble is proposed in this study. The overall flowchart is shown in Fig. 6, and the main steps are listed as follows:

(1) Collect the signals under different machine health conditions for model training. These machine health conditions are predetermined,

and thus are referred as seen classes. The signals are then divided into segments with a fixed length. Each segment is denoted as a sample.

(2) Construct the individual DBL, and combine them as the deep ensembles. It should be noted that different network architectures and learning mechanisms are considered in the used networks described in Section 2.2, to induce diversity. The members in the deep ensembles are not limited to the networks in this study. Use the samples to independently train the DBLs in parallel, and average their predictions as the final output of deep ensembles.

(3) After training, the deep ensembles is deployed to monitor the machine health condition in practical scenarios. Both the ID data from seen machine health conditions and OOD data from the unseen faults could be monitored. According to Eqs. (3) and (4), the predicted probability of machine health condition and the corresponding predictive uncertainty are calculated by deep ensembles.

(4) The predictive uncertainty, that is lower than the threshold, suggests the corresponding sample is likely in distribution and is from the seen machine health conditions. The predicted probability of machine health condition can be treated as the trustworthy fault diagnosis results. When the predictive uncertainty is higher than the threshold, the sample is identified as the OOD samples, such as unseen fault samples. And, the untrustworthy results are examined by the human expert.

To test the proposed approach, accuracy is effective to evaluate the percentage of the correctly predicted samples:

$$Accuracy = \frac{|x_i : x_i \in D \wedge \hat{y}_i = y_i|}{|x_i : x_i \in D|} \times 100\%, \tag{8}$$

where y_i is the true category of testing sample x_i, and $\hat{y}_i$ is the predicted label by the fault diagnosis model, and D is the testing set.

Specially, the proposed fault diagnosis method need to handle the mixture of both ID and OOD data in this study. The detection rate of untrustworthy results of OOD samples is expected to be as high as possible in safety-critical applications, while the false alarm rate with respect to ID samples should be as low as possible, since the additional human intervention is time-consuming and labor-intensive. Consequently, another two indicators are used for verification, namely the false alarm rate (FAR) and the missing alarm rate (MAR). Note that the FAR and MAR are evaluated with respect to the detection rate of ID and OOD samples. Their definitions are given as follows:

$$FAR = \frac{FP}{FP + TN} \times 100\%, \tag{9}$$

$$MAR = \frac{FN}{TP + FN} \times 100\%, \tag{10}$$

where TP, FP, FN and TN are the numbers of true positives, false positives, false negatives and true negatives, respectively, the positives denote the OOD samples and the negatives represent the ID samples.

3. Experimental analysis

3.1. Case study 1: Wind turbine fault diagnosis

3.1.1. Experiments description

The first case study uses the original vibration signals collected from the wind turbine test rig to create the fault dataset. Fig. 7(a) presents a schematic diagram of the wind turbine test rig. A wind tunnel capable of generating wind sources at a maximum wind speed of 15 m/s is used to drive the wind turbine in order to simulate real-world operational circumstances. In this case study, the wind speed is set around 10 m/s and 11 m/s. The vibration acceleration signals (sampling frequency: 20 kHz) at front and back bearing pedestals are used to monitor the health condition of the wind turbine. After data preparation, the original signals are divided into samples with 8192 data points.

As the failure easily occurs in rolling bearing, the data under health status (H) and three bearing faults are used to trained the fault diagnosis model. The three bearing faults, i.e., outer race fault (OF), inner race fault (IF) and roller fault (RF) are also referred as seen/known faults in this case. In real applications, the trained fault diagnosis model is likely to deal with the faults out of the known scope, referred as unseen/unknown faults, due to the complex machine structure, failure mechanism as well as operating conditions. Two unseen faults are considered in this case. One is about the fault in the mechanical system of transmission chain, that is, the loose fault of back bearing pedestal (LF). The other is related to the wind wheel, namely, aerodynamic asymmetry (AA), caused by the variation in airfoil of blades. The details of OF, IF, RF and LF are shown in Fig. 7(b). Three diagnostic experiments are designed, whose details are described in Table 1. The training and testing samples among different health classes are assumed to be balanced in this study. Note that the AA and LF are regarded as one label, i.e., unseen fault, in experiment 3, as no additional diagnosis knowledge can be used to determine the specific fault type for OOD samples, and the goal of our approach is to automatically sign the unknowns during diagnosis and resort to human experts to avoid the untrustworthy decision-making [25].

3.1.2. Trustworthy diagnosis results and analysis

The results of three wind turbine fault diagnosis experiments are given in this section, and the analysis is mainly twofold: (1) demonstrate the effectiveness of the proposed approach to identify unseen faults and give trustworthy diagnostic decisions with respect to conventional diagnosis in a data-driven framework, (2) demonstrate the superiority of the deep ensembles with respect to individual deep models in the aspect of quantifying the uncertainty and detecting the samples from unseen faults.

First, the diagnostic accuracies, the FARs and the MARs are shown in Tables 2 and 3, respectively. To be more specific, the multi-class confusion matrices of the conventional diagnosis approach in experiment 3 are shown in Fig. 8. All the individual DBLs and the deep ensembles can accurately diagnose the ID samples (labels 0–3), while fully misdiagnose the OOD samples of unseen faults. As consequence, the accuracies of diagnosing the mixture testing data are close to 80% in experiment 1 and experiment 2, and 66.7% in experiment 3. Obviously, when following the traditional data-driven diagnosis framework, these models are all learnt with the closed-set data, that is, the data from health and three bearing faults. The samples of unseen fault are assumed to be the ID data and are blindly classified into the known labels (0–3) by these trained model. It is a significant challenge from unseen faults, to hinder the trustworthy decision-making. The necessity of the proposed trustworthy fault diagnosis approach is clearly illustrated.

Second, by comparing the results in Tables 2 and 3, it is obvious the diagnostic performances for most of models are improved with the proposed approach. Specially, in the three diagnostic experiments, the deep ensembles with trustworthy analysis achieved 8.5%, 8.5% and 10.4% improvements of accuracy. The corresponding multi-class confusion matrices of the different models in experiment 3 are presented in Fig. 9. It is clear to find the identification rates of unseen faults (AA and LF) are improved accordingly for the six different models. Note that the identification rates of unseen faults in Fig. 9 correspond to the MARs in Table 3. In safety-critical applications, particular attention needs to be devoted to the trustworthiness of diagnosis results. Therefore, the MARs of the unseen faults and even unknown OOD samples, should be maintained as low as possible. Once the unseen fault occurs, the diagnostic model should accurately identify it and issue the warning for potential expert intervention. In this study, the IQR proximity rule is utilized to determine the uncertainty threshold within the ID set. And part of ID testing samples with higher uncertainty are classified into the unseen fault, leading to the FARs around 15.0% (as listed in Table 3). Overall, the proposed trustworthy fault diagnosis approach with deep

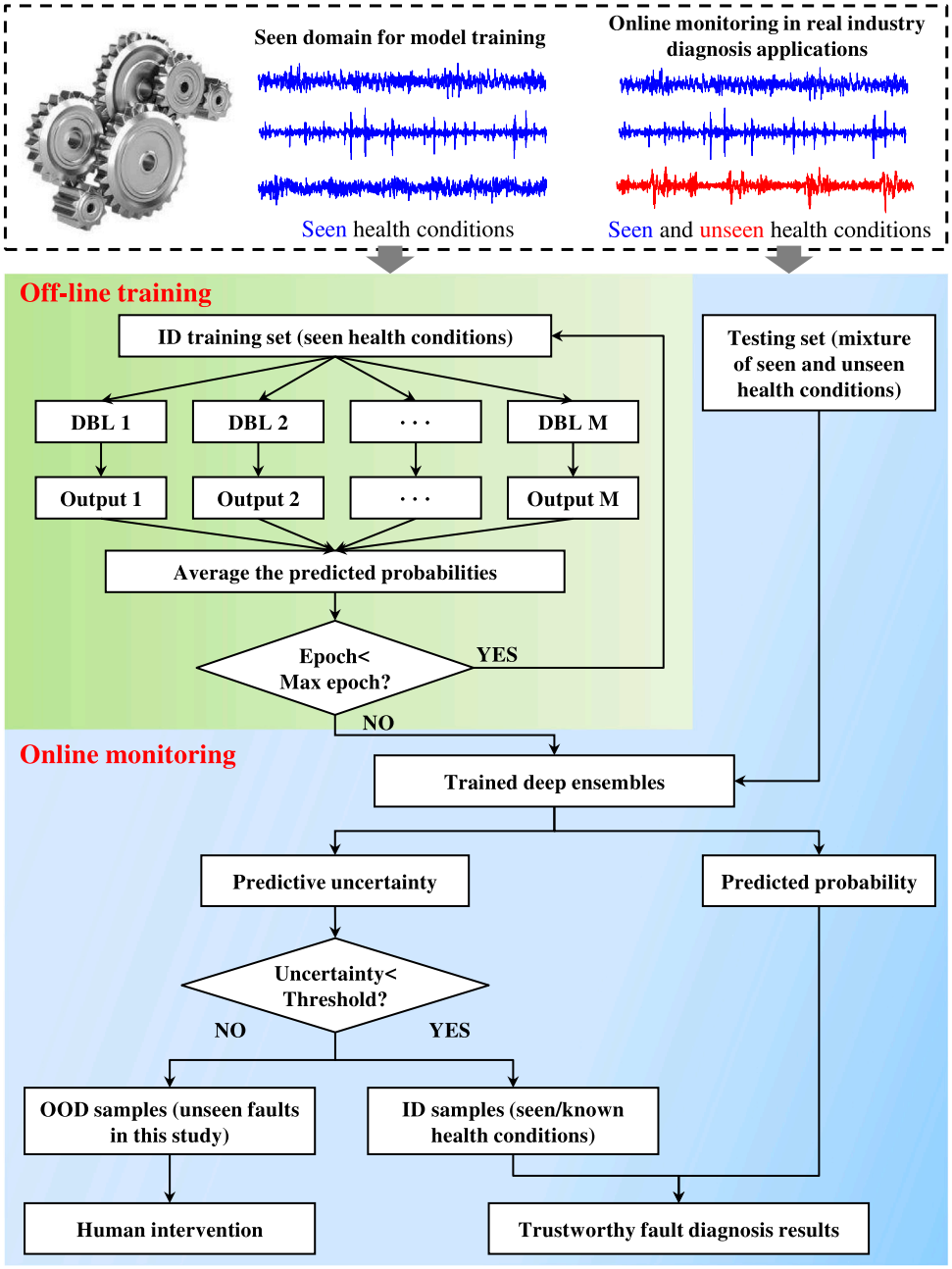

Fig. 6. The flowchart of the proposed trustworthy fault diagnosis approach.

Table 1
Descriptions of wind turbine diagnosis experiments.

Experiment	Seen health conditions	Unseen faults	Number of training samples	Number of testing samples
No. 1	H, OF, IF and RF labels 0, 1, 2, 3	AA label 4	4 × 1000[a]	5 × 250[b]
No. 2	H, OF, IF and RF labels 0, 1, 2, 3	LF label 4	4 × 1000	5 × 250
No. 3	H, OF, IF and RF labels 0, 1, 2, 3	AA and LF label 4	4 × 1000	6 × 250

[a]There are 1000 sample of each seen health condition for the training of diagnostic model.
[b]The testing samples is a mixture from both seen and unseen classes. There are 250 testing samples for each health condition.

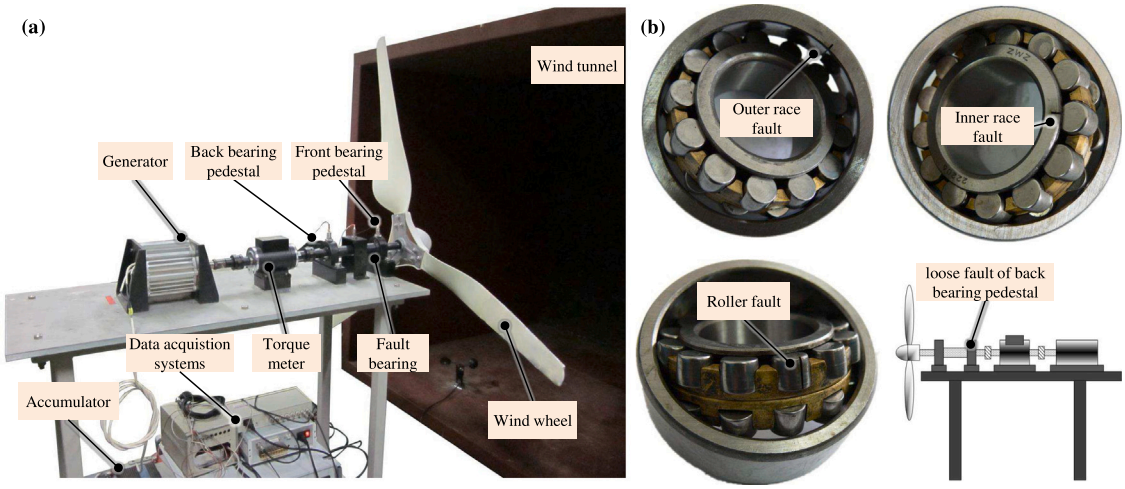

Fig. 7. The direct-driven wind turbine test rig. (a) Its structure, (b) Details of the faults.

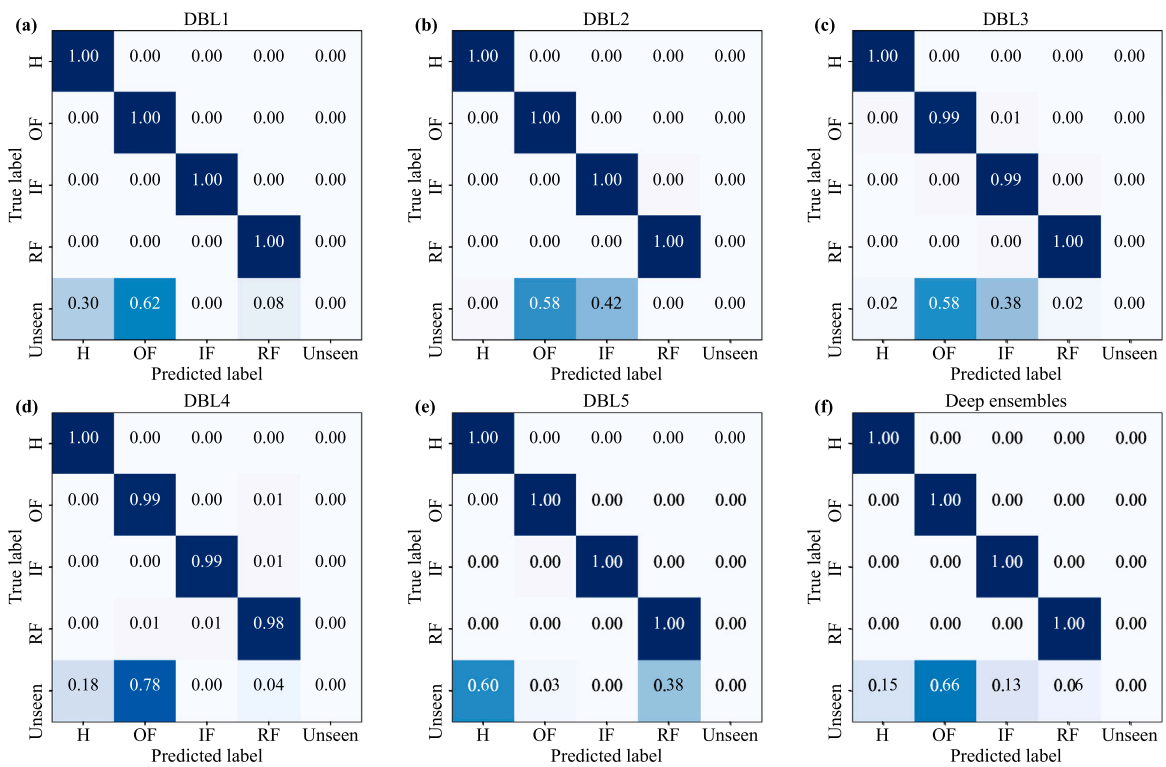

Fig. 8. The confusion matrices of different models in experiment 3 (**Conventional diagnosis strategy without trustworthy analysis**).

Table 2
Diagnostic results (%) of experiments 1–3 without trustworthy analysis.[a]

Methods	Experiment 1			Experiment 2			Experiment 3		
	Accuracy	FAR	MAR	Accuracy	FAR	MAR	Accuracy	FAR	MAR
DBL1[b]	79.9	n[c]	n	79.9	n	n	66.6	n	n
DBL2	79.9	n	n	79.9	n	n	66.6	n	n
DBL3	79.4	n	n	79.4	n	n	66.2	n	n
DBL4	79.4	n	n	79.4	n	n	66.1	n	n
DBL5	79.9	n	n	79.9	n	n	66.6	n	n
Deep ensembles	80.0	n	n	80.0	n	n	66.7	n	n

[a]**Conventional diagnosis approach without trustworthy analysis:** The diagnostic accuracy of the mixture (ID data of four seen classes and OOD data of unseen faults) is directly given by the trained model.

[b]The described five DBLs in Section 2.2, namely CNN with three convolutional layers, CNN with four convolutional layers, CNN with five convolutional layers, ResNet, and inception network, are marked as DBL1, DBL2, DBL3, DBL4 and DBL5, respectively.

[c]In the context of conventional fault diagnosis, no FARs and MARs can be reported for the trustworthiness of diagnosis results.

Table 3
Diagnostic results (%) of experiments 1–3 considering trustworthy analysis.[a]

Methods	Experiment 1			Experiment 2			Experiment 3		
	Accuracy	FAR	MAR	Accuracy	FAR	MAR	Accuracy	FAR	MAR
DBL1	82.6	10.8	43.6	86.2	10.8	26.0	80.9	10.8	35.8
DBL2	81.7	14.2	34.8	73.8	14.2	74.4	72.4	14.2	54.4
DBL3	86.4	10.2	27.2	78.7	10.2	66.0	77.3	10.2	47.6
DBL4	81.8	11.8	43.6	87.6	11.8	14.4	81.4	11.8	32.0
DBL5	90.6	8.9	11.2	85.0	8.9	39.6	85.8	8.9	24.8
Deep ensembles	92.2	9.7	0.0	92.2	9.7	0.0	93.5	9.7	0.0

[a]**Proposed diagnosis approach with trustworthy analysis:** The OOD samples of unseen faults are detected by the uncertainty-aware models.

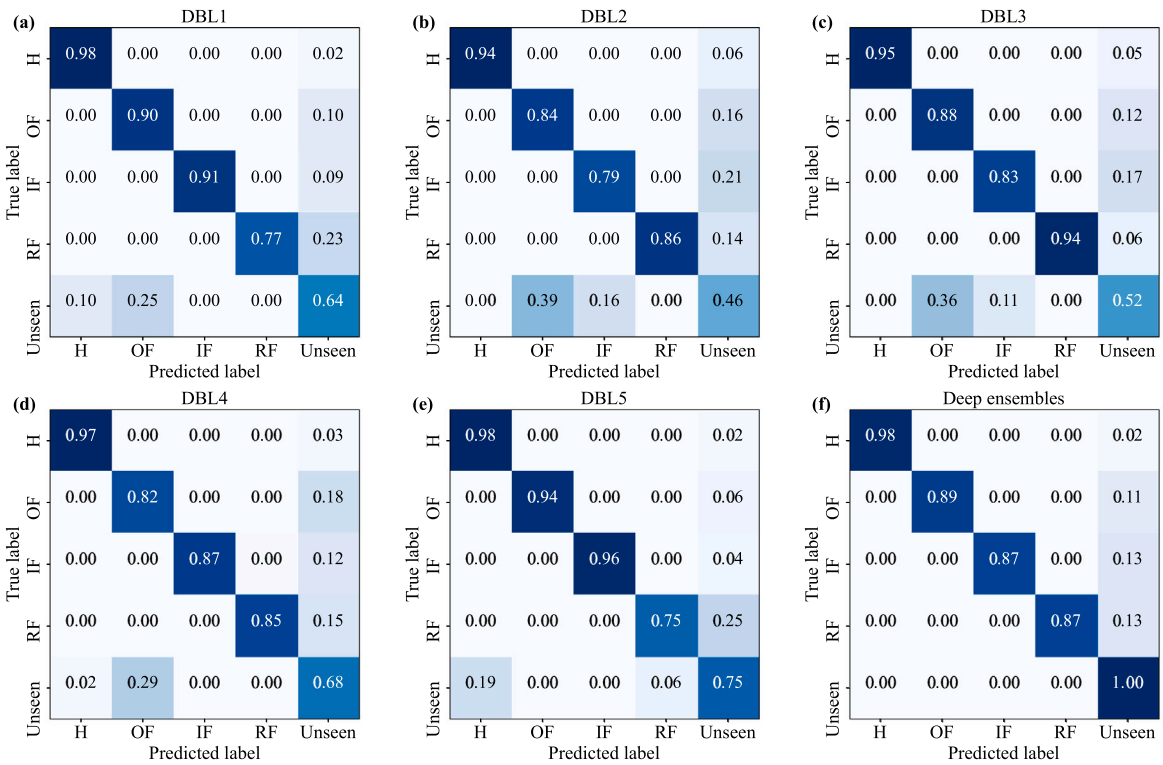

Fig. 9. The confusion matrices of different models in experiment 3 (**Proposed diagnosis strategy with trustworthy analysis**).

ensembles can effectively improve the diagnostic accuracy and entirely isolate the OOD samples of unseen faults.

Third, through a series of comparisons between individual deep learning models and deep ensembles (as shown in Table 3), the advantages of deep ensembles are prominent. All the samples from unseen faults are detected by deep ensembles, achieving the 0.0% MARs in the three experiments, whereas most of the MARs for the five DBLs are over 20.0%. As consequence, the deep ensembles also has the highest diagnostic accuracies. Since the unseen fault samples are isolated by the uncertainty-guided OOD detection strategy, the presented deep ensembles possesses superior property to understand what are unknowns for the trained model, and significantly promotes the uncertainty estimates. More intuitive explanations will be given in Section 3.1.3.

3.1.3. Comparisons of uncertainty distribution of ID and OOD data

In our proposed approach, the predictive uncertainty plays an key role in accurately detecting the unseen fault samples. Nevertheless, the individual deep learning models tend to make overconfident results on the OOD samples, and show the poor performance for uncertainty estimation. The deep ensembles further exploits the disagreement of all DBLs, and boosts the uncertainty quality. To confirm this point, Fig. 10 presents the softmax outputs of the five individual DBLs (DBL1–DBL5) and the deep ensembles on an ID sample and an OOD sample. It is clear that all the six models place nearly all of probability mass on the true target for the health sample, while give different decisions on the unseen fault sample. As the unseen fault is out of the known health conditions, it is unavailable to know the true softmax distribution, and we anticipate that the softmax output of unseen fault samples should be more uniform on different known class, so as to reflect the models' unknowns during diagnosis. Compared to the overconfident and wrong diagnosis of DBLs, the deep ensembles achieved desirable results on unseen fault sample, which will be more suitable to calculate the uncertainty measure and transmit the trustworthiness of diagnostic results.

Furthermore, taking the experiment 3 as an example, the uncertainty distributions of all the ID and OOD testing samples are shown in Fig. 11. For the DBLs, there are large overlapping areas between ID and OOD samples. It is difficult to separate the OOD samples, whose uncertainty is lower than the threshold, from the ID samples. For the proposed deep ensembles, the uncertainty of OOD samples is much higher than the ones of ID samples, suggesting these OOD samples can be easily detected and classified into unseen fault. These observations effectively illustrate the advantages of deep ensembles in terms of finding the unknowns and capturing more efficient predictive uncertainty.

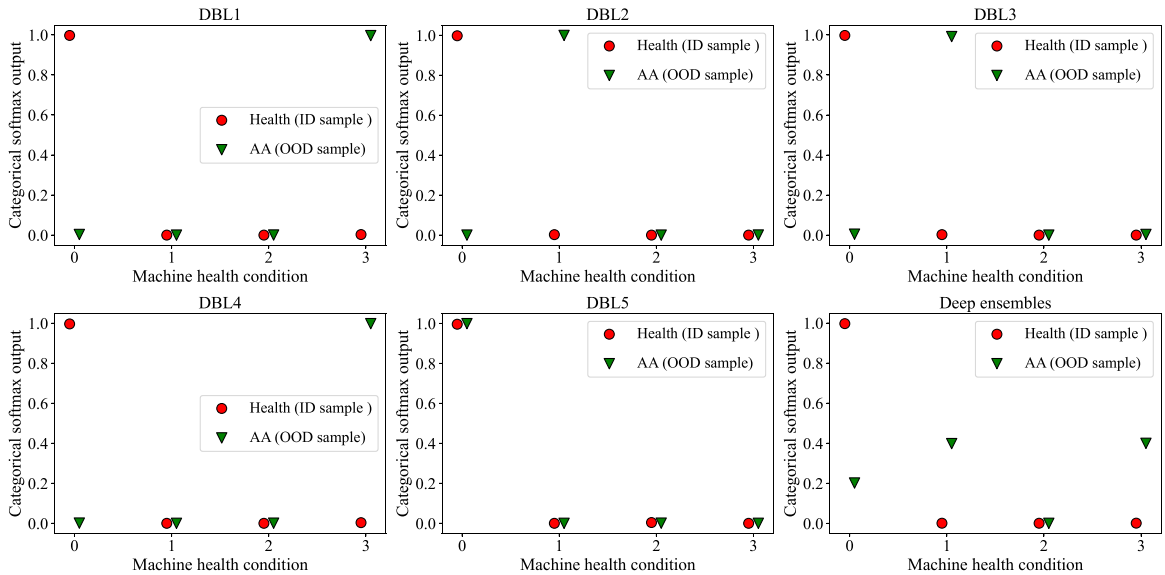

Fig. 10. The softmax predictions on an ID sample (health) and an OOD sample (unseen fault: AA) in case study 1.

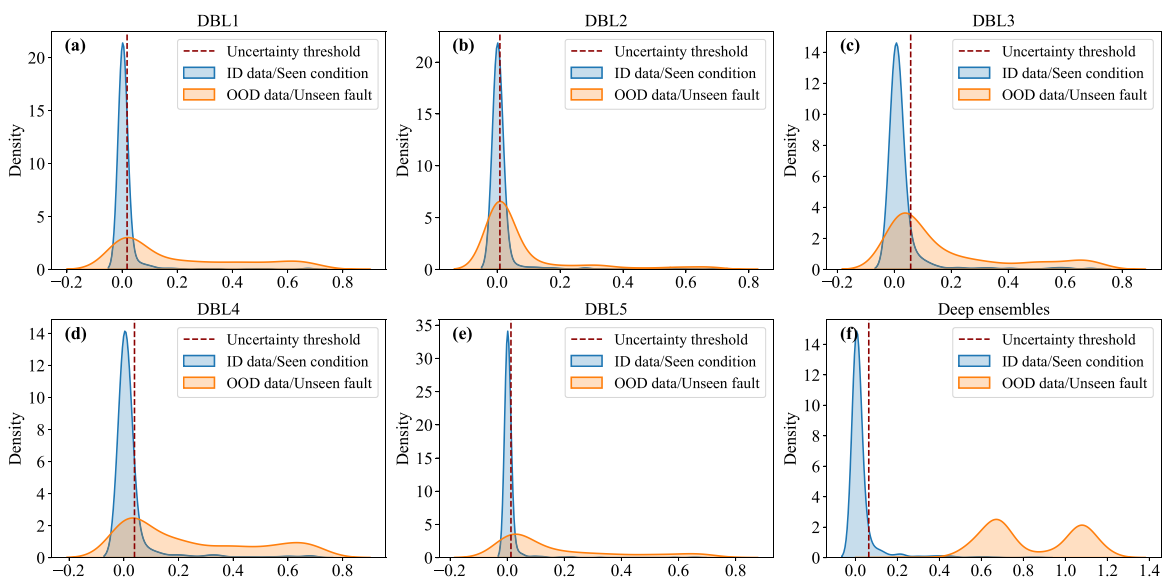

Fig. 11. The uncertainty distributions of ID (seen health condition) and OOD (unseen fault) testing samples in experiment 3.

3.2. Case study 2: Gearbox fault diagnosis

3.2.1. Experiments description

The second case study demonstrates the proposed method in a HS-200 planetary gearbox fault dataset. The gearbox is a single-stage planetary gear, whose reduction ratio is 3.57. Part of its structure is shown in Fig. 12(a). The motor is used to drive the gearbox and the output end is connected to a wind wheel as the load. The rotating speeds of input end ranges from 20 Hz to 40 Hz. The gearbox vibration signal is collected by the accelerator installed on housing. The sampling frequency is 20 kHz. After data pre-processing, each sample contains 4096 data points.

In the designed experiments, the health status (H) and four planetary gear faults are treated as seen/known faults, while two sun gear faults are unseen during model training. The planetary gear faults include the tooth cracks (crack depth: 1/8 dedendum, 1/4 dedendum and 1/2 dedendum, respectively, denoted as PC1, PC2, PC3) and the tooth broken (position: 1/3 tooth depth, denoted as PB). The sun gear faults include the tooth crack with the depth of 1/8 dedendum (SC), and the tooth broken at the position of 1/3 tooth depth (SB). Fig. 12(b) presents the two damaged planetary gears (PC1 and PB) and two damaged sun gears (SC and SB). The details of the three diagnostic experiments are listed in Table 4.

3.2.2. Trustworthy diagnosis results and analysis

Similar to the analysis in the case study 1, the diagnostic results without/with trustworthy analysis are given in Tables 5 and 6, respectively. Taking the example 6 as an example, the multi-class confusion matrices without/with trustworthy analysis are shown in Figs. 13 and 14, respectively. By comparing Figs. 13 and 14, it is found that a number of OOD samples of unseen fault in the mixed testing set are detected after considering the trustworthy analysis. And thus, most of models achieve higher diagnostic accuracies with the trustworthy analysis. In particular, as listed in Table 6, the MAR of the presented deep ensembles is only 3.3% in experiment 6. The corresponding diagnosis accuracy for the OOD samples of unseen fault is 97.0% in Fig. 14. In the Table 6, it is also clear that the deep ensembles always achieves higher accuracies than individual DBLs, and lower MARs than individual

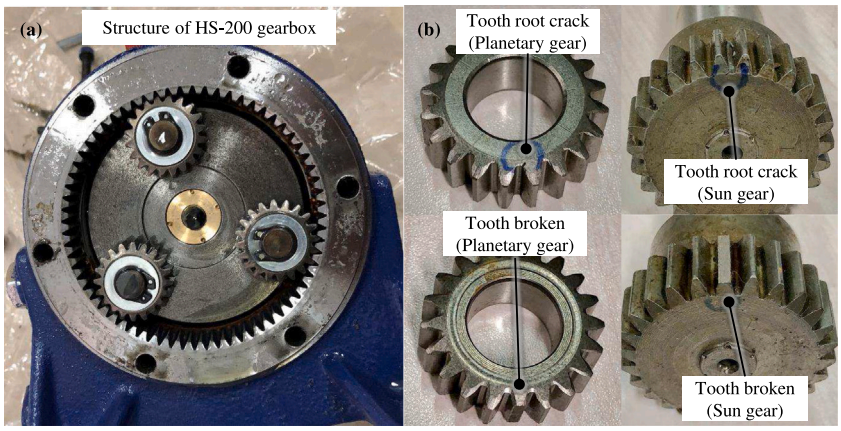

Fig. 12. The HS-200 planetary gearbox. (a) Its structure, (b) Gear fault types.

Table 4
Descriptions of gearbox diagnosis experiments.

Experiment	Seen health conditions	Unseen faults	Number of training samples	Number of testing samples
No. 4	H, TC1, TC2, TC3 and TB labels 0, 1, 2, 3, 4	SC label 5	5 × 1000	6 × 200
No. 5	H, TC1, TC2, TC3 and TB labels 0, 1, 2, 3, 4	SB label 5	5 × 1000	6 × 200
No. 6	H, TC1, TC2, TC3 and TB labels 0, 1, 2, 3, 4	SC and SB label 5	5 × 1000	7 × 200

Table 5
Diagnostic results (%) of experiments 4–6 without trustworthy analysis.

Methods	Experiment 4			Experiment 5			Experiment 6		
	Accuracy	FAR	MAR	Accuracy	FAR	MAR	Accuracy	FAR	MAR
DBL1	83.0	n	n	83.0	n	n	71.1	n	n
DBL2	83.0	n	n	83.0	n	n	71.1	n	n
DBL3	82.6	n	n	82.6	n	n	70.8	n	n
DBL4	82.1	n	n	82.1	n	n	70.4	n	n
DBL5	83.0	n	n	83.0	n	n	71.1	n	n
Deep ensembles	83.2	n	n	83.2	n	n	71.3	n	n

Table 6
Diagnostic results (%) of experiments 4–6 considering trustworthy analysis.

Methods	Experiment 4			Experiment 5			Experiment 6		
	Accuracy	FAR	MAR	Accuracy	FAR	MAR	Accuracy	FAR	MAR
DBL1	84.1	11.3	39.0	85.7	11.3	29.5	82.0	11.3	34.8
DBL2	81.6	13.1	44.5	85.4	13.1	21.5	80.1	13.1	36.8
DBL3	69.8	16.9	96.5	78.3	16.9	45.5	67.3	16.9	72.0
DBL4	69.7	19.3	85.0	79.4	19.3	26.5	68.6	19.3	61.3
DBL5	74.2	15.2	79.0	84.4	15.2	17.5	74.3	15.2	52.0
Deep ensembles	89.9	12.9	3.25	89.2	12.9	0.5	89.9	12.9	3.3

DBLs. These observations further demonstrate the effectiveness of the proposed trustworthy approach with deep ensembles when handling the OOD fault diagnosis problems.

3.2.3. Comparisons of uncertainty distribution of ID and OOD data

To intuitively illustrate the principle of proposed trustworthy fault diagnosis method as well as the superiority of deep ensembles, the softmax outputs of the five individual DBLs (DBL1–DBL5) and the deep ensembles on an ID sample and an OOD sample are presented in Fig. 15. Similarly, the individual DBL tends to give overconfident results for the OOD sample, since most deep neural networks are inherently deterministic [27]. The presented deep ensembles is better at quantifying the uncertainty. Moreover, the uncertainty distributions of both ID samples (seen health conditions) and OOD samples (unseen faults) are shown in Fig. 16. In the DBLs, the large overlappings of ID and OOD samples indicate that the individual deep learning models cannot separate the OOD samples from the mixture of seen and unseen health conditions by using the uncertainty measure. The reason is that all these DBLs lack the ability to quantify the uncertainty, and the output predictions of the DBLs are deterministic and easily overconfident. Meanwhile, it is clear to observe that the uncertainty of OOD samples significantly increases for the deep ensembles. The two uncertainty distributions of ID and OOD data are separable. All these results suggest that the presented deep ensembles is suitable for perceiving the unknowns and detecting the OOD samples of unseen faults towards the trustworthy fault diagnosis.

3.3. Limitations of the proposed approach

Despite the fact that the proposed approach outperforms the standard deep learning model and guarantees the trustable decisions for

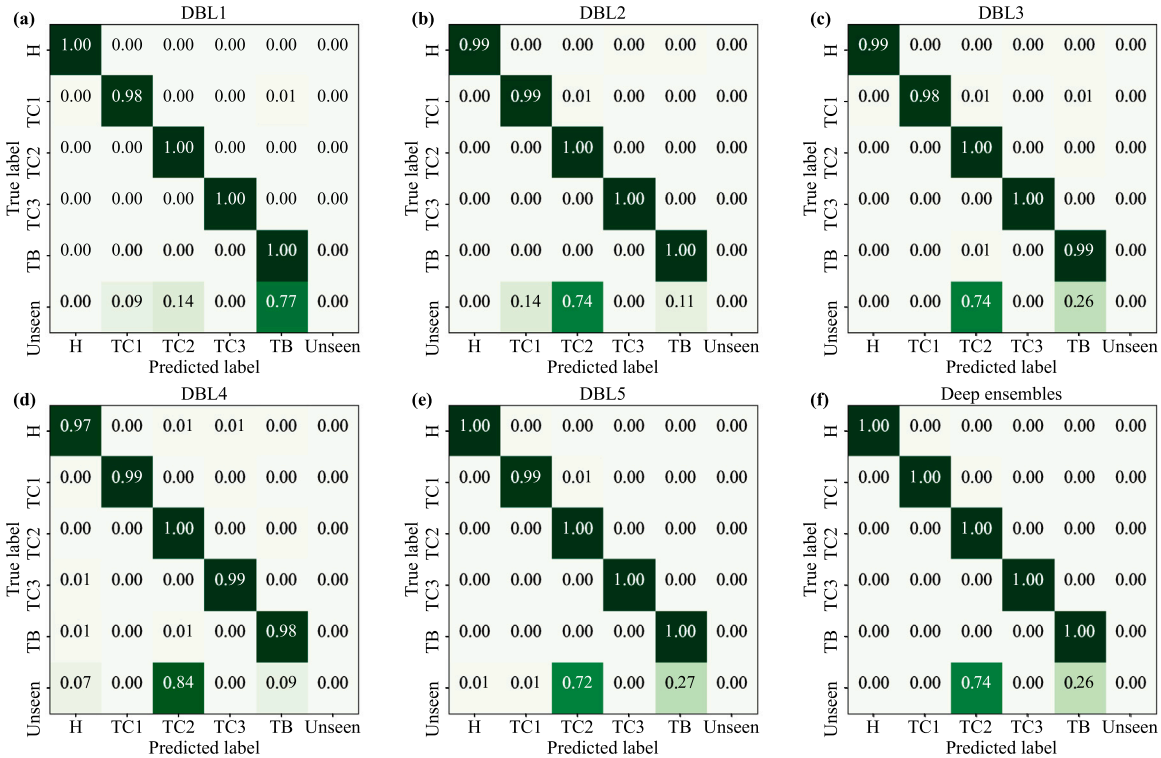

Fig. 13. The confusion matrices of different models in experiment 6 (**Conventional diagnosis strategy without trustworthy analysis**).

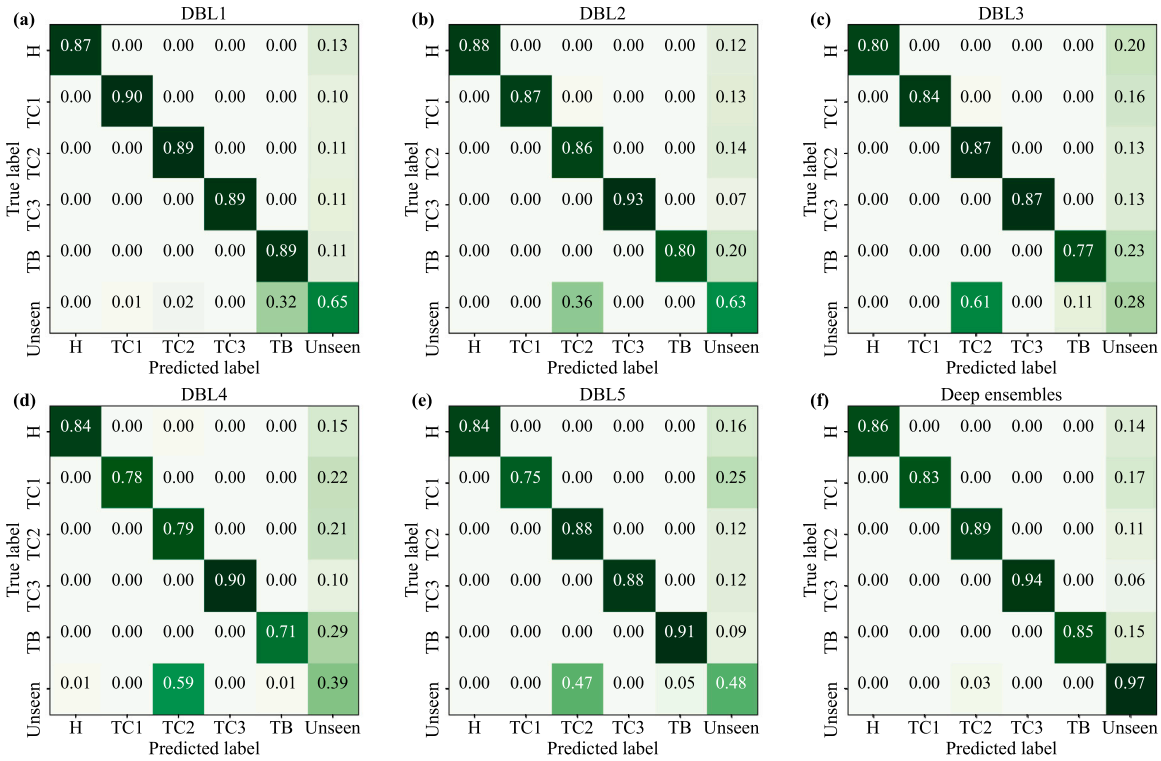

Fig. 14. The confusion matrices of different models in experiment 6 (**Proposed diagnosis strategy with trustworthy analysis**).

machinery fault diagnosis, some limitations are also necessary to be clarified.

(1) Attribution of OOD samples. In the experiments, the effectively detected OOD samples are assumed to be attributed to unseen faults. In addition to the unseen faults, OOD samples may be caused by other reasons such as unseen working conditions and unseen noise environments. The proposed approach cannot identify the OOD samples from unseen faults, unseen working conditions or unseen noise

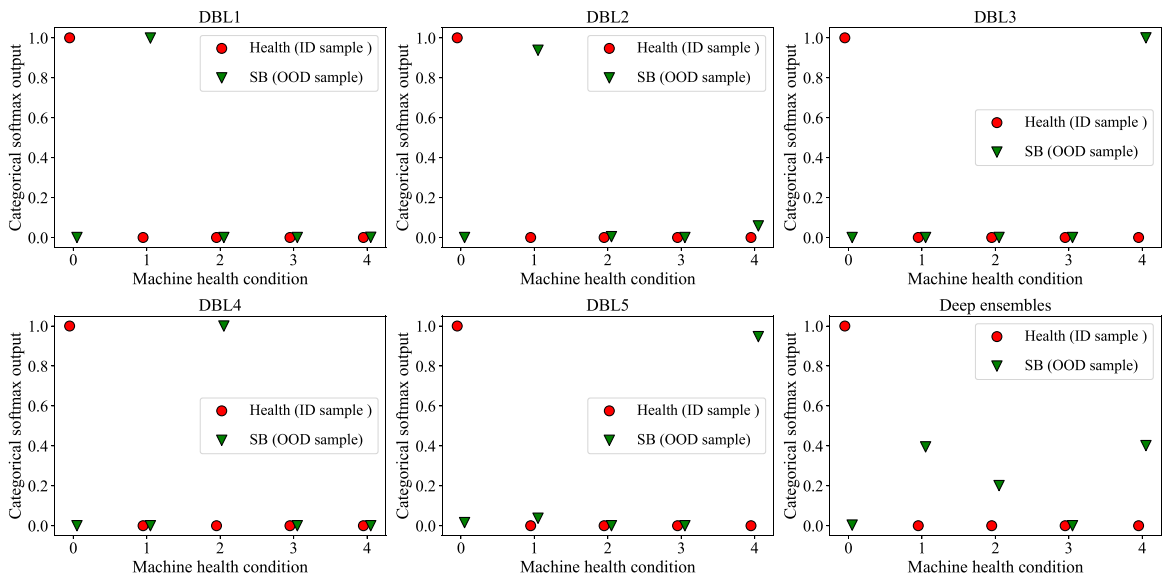

Fig. 15. The softmax predictions on an ID sample (health) and an OOD sample (unseen fault: SB) in case study 2.

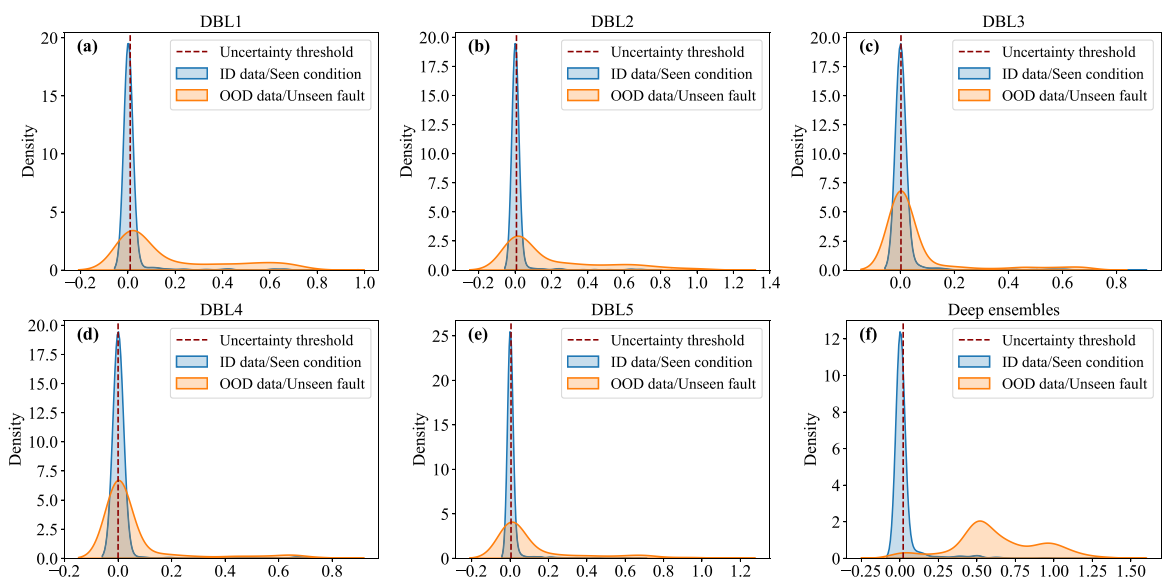

Fig. 16. The uncertainty distributions of ID (seen health condition) and OOD (unseen fault) testing samples in experiment 6.

environments, but only utilize the uncertainty to detect the unknowns during model's decision marking. The attribution of OOD samples will be conducive to the understanding of models' alarms for human intervention, which is still an unsolved problem in this study.

(2) Diversity for deep base learners. The diversity among different deep base learners is very important for accurately quantifying the uncertainty behind diagnosis decisions. Different model architectures are adopted to induce the diversity in this study. Other approaches, such as different activation functions and random shuffling of training data, also hold the potential to diversify base learners. Utilizing the optimal diversity generation mechanism to construct an uncertainty-aware ensemble system is a challenging problem. The design of the deep ensemble still relies on prior knowledge and experience.

4. Conclusions

In this paper, an OOD detection-assisted trustworthy machine fault diagnosis approach is proposed with uncertainty-aware deep ensembles. The proposed approach is capable of perceiving the predictive uncertainty during diagnosis, and thus know what are the unknowns for the trained model and when the diagnostic decisions can be trusted. Moreover, compared with the individual DBLs, the presented deep ensembles can capture more accurately uncertainty estimates in terms of aleatoric uncertainty and epistemic uncertainty. The effectiveness and superiority of the proposed approach is verified in two case studies, including a wind turbine fault diagnosis case and a gearbox fault diagnosis case. The experimental results demonstrate that the proposed approach is promising to handle the unseen faults problem, and achieve the trustworthy fault diagnosis results.

In the future study, another open question we are interested in is whether the known faults with the unseen severity degrees could be identified as untrustworthy OOD samples. It is a significant question because the model trained on the specific fault degrees need to monitor the early or progressed faults in practical applications. How to improve the robustness of proposed trustworthy fault diagnosis approach with respect to the discrepant yet ID samples will be investigated.

CRediT authorship contribution statement

Te Han: Writing – original draft, Software, Methodology, Formal analysis, Data curation, Conceptualization, Writing – review & editing. **Yan-Fu Li:** Writing – review & editing, Validation, Supervision, Resources, Investigation, Formal analysis, Funding acquisition, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

[1] Zio E. Prognostics and health management (PHM): Where are we and where do we (need to) go in theory and practice. Reliab Eng Syst Saf 2022;218:108119.
[2] Hu Y, Miao X, Si Y, Pan E, Zio E. Prognostics and health management: A review from the perspectives of design, development and decision. Reliab Eng Syst Saf 2022;217:108063.
[3] Li X, Zhong X, Shao H, Han T, Shen C. Multi-sensor gearbox fault diagnosis by using feature-fusion covariance matrix and multi-Riemannian kernel ridge regression. Reliab Eng Syst Saf 2021;216:108018.
[4] Xiang S, Qin Y, Liu F, Gryllias K. Automatic multi-differential deep learning and its application to machine remaining useful life prediction. Reliab Eng Syst Saf 2022;223:108531.
[5] Han T, Liu C, Wu L, Sarkar S, Jiang D. An adaptive spatiotemporal feature learning approach for fault diagnosis in complex systems. Mech Syst Signal Process 2019;117:170–87.
[6] Wang W, Lei Y, Yan T, Li N, Nandi A. Residual convolution long short-term memory network for machines remaining useful life prediction and uncertainty quantification. J Dyn Monitor Diagnost 2021;1(1):2–8.
[7] Chen Z, He G, Li J, Liao Y, Gryllias K, Li W. Domain adversarial transfer network for cross-domain fault diagnosis of rotary machinery. IEEE Trans Instrum Meas 2020;69(11):8702–12.
[8] Zhao K, Jiang H, Liu C, Wang Y, Zhu K. A new data generation approach with modified wasserstein auto-encoder for rotating machinery fault diagnosis with limited fault data. Knowl-Based Syst 2022;238:107892.
[9] Han T, Li Y-F, Qian M. A hybrid generalization network for intelligent fault diagnosis of rotating machinery under unseen working conditions. IEEE Trans Instrum Meas 2021;70:1–11.
[10] Shim J-H, Lee K, Ha J-I. Fault diagnosis of inverter current sensor using artificial neural network considering out-of-distribution. In: 2021 IEEE 12th energy conversion congress exposition - asia (ECCE-Asia). 2021, p. 1520–5. http://dx.doi.org/10.1109/ECCE-Asia49820.2021.9479085.
[11] Zhou T, Droguett EL, Mosleh A, Chan FT. An uncertainty-informed framework for trustworthy fault diagnosis in safety-critical applications. 2021, arXiv preprint arXiv:2111.00874.
[12] Yang B, Xu S, Lei Y, Lee C-G, Stewart E, Roberts C. Multi-source transfer learning network to complement knowledge for intelligent diagnosis of machines with unseen faults. Mech Syst Signal Process 2022;162:108095.
[13] He A, Jin X. Deep variational autoencoder classifier for intelligent fault diagnosis adaptive to unseen fault categories. IEEE Trans Reliab 2021;70(4):1581–95.
[14] He Z, Shao H, Jing L, Cheng J, Yang Y. Transfer fault diagnosis of bearing installed in different machines using enhanced deep auto-encoder. Measurement 2020;152:107393.
[15] Han T, Liu C, Yang W, Jiang D. Deep transfer network with joint distribution adaptation: A new intelligent fault diagnosis framework for industry application. ISA Trans 2020;97:269–81.
[16] Qin Y, Yao Q, Wang Y, Mao Y. Parameter sharing adversarial domain adaptation networks for fault transfer diagnosis of planetary gearboxes. Mech Syst Signal Process 2021;160:107936.
[17] Wang X, Shen C, Xia M, Wang D, Zhu J, Zhu Z. Multi-scale deep intra-class transfer learning for bearing fault diagnosis. Reliab Eng Syst Saf 2020;202:107050.
[18] Lee J, Kim M, Ko JU, Jung JH, Sun KH, Youn BD. Asymmetric inter-intra domain alignments (AIIDA) method for intelligent fault diagnosis of rotating machinery. Reliab Eng Syst Saf 2022;218:108186.
[19] Park C, Awadalla A, Kohno T, Patel S. Reliable and trustworthy machine learning for health using dataset shift detection. Adv Neural Inf Process Syst 2021;34.
[20] Nitsch J, Itkina M, Senanayake R, Nieto J, Schmidt M, Siegwart R, et al. Out-of-distribution detection for automotive perception. In: 2021 IEEE international intelligent transportation systems conference (ITSC). IEEE; 2021, p. 2938–43.
[21] Linmans J, van der Laak J, Litjens G. Efficient out-of-distribution detection in digital pathology using multi-head convolutional neural networks. In: MIDL. 2020, p. 465–78.
[22] Cao T, Huang C-W, Hui DY-T, Cohen JP. A benchmark of medical out of distribution detection. 2020, arXiv preprint arXiv:2007.04250.
[23] Zadorozhny K, Thoral P, Elbers P, Cinà G. Out-of-distribution detection for medical applications: Guidelines for practical evaluation. 2021, arXiv preprint arXiv:2109.14885.
[24] Pimentel MA, Clifton DA, Clifton L, Tarassenko L. A review of novelty detection. Signal Process 2014;99:215–49.
[25] Yang J, Zhou K, Li Y, Liu Z. Generalized out-of-distribution detection: A survey. 2021, arXiv preprint arXiv:2110.11334.
[26] Chai LR. Uncertainty estimation in bayesian neural networks and links to interpretability. (Master's Thesis), University of Cambridge; 2018.
[27] Nguyen A, Yosinski J, Clune J. Deep neural networks are easily fooled: High confidence predictions for unrecognizable images. In: Proceedings of the IEEE conference on computer vision and pattern recognition, 2015. p. 427–36.
[28] Gawlikowski J, Tassi CRN, Ali M, Lee J, Humt M, Feng J, et al. A survey of uncertainty in deep neural networks. 2021, arXiv preprint arXiv:2107.03342.
[29] Lakshminarayanan B, Pritzel A, Blundell C. Simple and scalable predictive uncertainty estimation using deep ensembles. Adv Neural Inf Process Syst 2017;30.
[30] Ţifrea A, Stavarache E, Yang F. Novelty detection using ensembles with regularized disagreement. 2020, arXiv preprint arXiv:2012.05825.
[31] Zhang W, Peng G, Li C, Chen Y, Zhang Z. A new deep learning model for fault diagnosis with good anti-noise and domain adaptation ability on raw vibration signals. Sensors 2017;17(2):425.
[32] Han T, Liu C, Yang W, Jiang D. A novel adversarial learning framework in deep convolutional neural network for intelligent diagnosis of mechanical faults. Knowl-Based Syst 2019;165:474–87.
[33] Wang H, Liu Z, Ai T. Long-range dependencies learning based on non-local 1D-convolutional neural network for rolling bearing fault diagnosis. J Dyn Monitor Diagnost 2022. http://dx.doi.org/10.37965/jdmd.2022.53.
[34] Jiao J, Zhao M, Lin J, Liang K. A comprehensive review on convolutional neural network in machine fault diagnosis. Neurocomputing 2020;417:36–63.
[35] He K, Zhang X, Ren S, Sun J. Deep residual learning for image recognition. In: 2016 IEEE conference on computer vision and pattern recognition (CVPR), 2016. p. 770–8.
[36] Krizhevsky A, Sutskever I, Hinton GE. Imagenet classification with deep convolutional neural networks. Adv Neural Inf Process Syst 2012;25.
[37] Simonyan K, Zisserman A. Very deep convolutional networks for large-scale image recognition. 2014, arXiv preprint arXiv:1409.1556.
[38] Szegedy C, Ioffe S, Vanhoucke V, Alemi AA. Inception-v4, inception-ResNet and the impact of residual connections on learning. In: Proceedings of the thirty-first AAAI conference on artificial intelligence. AAAI'17, AAAI Press; 2017, p. 4278–84.
[39] Breiman L. Random forests. Mach Learn 2001;45(1):5–32.
[40] Li Y, Zio E. Uncertainty analysis of the adequacy assessment model of a distributed generation system. Renew Energy 2012;41:235–44.
[41] Li Y-F, Ding Y, Zio E. Random fuzzy extension of the universal generating function approach for the reliability assessment of multi-state systems under aleatory and epistemic uncertainties. IEEE Trans Reliab 2014;63(1):13–25.